

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-28 15:27 UTC

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Generated 2026-04-28 15:27 UTC

- Cycles recorded: **197**
- Time span: 113.8h (~1.73 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	174
FALSIFIED	14
BARRIER_HIT	5
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier analysis of boolean functions	3
Noncommutative geometry	2
Representation theory of symmetric groups	2
Ergodic Circuit Framework (COMM_COMPLEXITY)	2
Matroid Theory	2
Algebraic Topology	2
Tropical geometry	2
Random Matrix Theory	2
Combinatorial homotopy theory	1
Kazhdan-Lusztig theory of Hecke algebras	1
Combinatorial algebraic topology	1
Knot theory (quantum invariants)	1
Quadratic forms over finite fields (Grothendieck-Witt groups)	1
Clifford algebras over finite fields	1
Toric geometry (via Gröbner degenerations)	1

Field	Cycles
Arithmetic geometry (Tate-Shafarevich groups)	1
Geometric Langlands correspondence	1
Algebraic graph theory (chromatic polynomials)	1
Clifford algebras over real vector spaces	1
Algebraic cycles (Chow groups over finite fields)	1
Algebraic lattice theory (Möbius functions of flow lattices)	1
Directed algebraic topology (d-space homology)	1
Motivic integration (Denef-Loeser zeta functions)	1
Matroid theory (Clifford index of binary matroids)	1
Representation theory of finite groups (Frobenius-Schur indicator)	1
Categorification and knot Floer homology	1
Modular forms	1
Algebraic geometry (ideals in polynomial rings)	1

Last 15 cycles (chronological)

Time	Verdict	Title
2026-04-28 09:45 UTC	INCONCLUSIVE	Plünnecke-Ruzsa Doubling of Constraint Vectors
2026-04-28 10:22 UTC	INCONCLUSIVE	Lower-Bounds SoS D Kolmogorov Width of Lifted Functions Bounds MA^{cc}
2026-04-28 10:39 UTC	INCONCLUSIVE	Tensor-Amplified Asdim Forces Linear Communication
2026-04-28 10:48 UTC	INCONCLUSIVE	Blow-up on Inn Möbius Defect of NW Design
2026-04-28 11:19 UTC	INCONCLUSIVE	Lattice Bounds Parity Bias Sandpile-Group Order of Certificate Conflict Graph
2026-04-28 11:47 UTC	INCONCLUSIVE	Lower-Bounds Q Stanley-Reisner Projective Dimension Lower-Bounds
2026-04-28 12:19 UTC	INCONCLUSIVE	Monotone-KW Dep Lie-Stabilizer Dimension of Clause Cubic Bounds DPLL
2026-04-28 12:42 UTC	INCONCLUSIVE	Leaves Følner-Defect Floor Forces Logarithmic Communication
2026-04-28 12:48 UTC	INCONCLUSIVE	Overhead on Bipartite Token-Sliding Diameter Lower-Bounds
2026-04-28 13:27 UTC	INCONCLUSIVE	Monotone-KW Depth Magnus Level-2 Defect Bounds DNF_min via Truth-Table
2026-04-28 13:54 UTC	INCONCLUSIVE	Inversions Monomer-Dimer Entropy Lower-Bounds Tseitin Resolution Width

Time	Verdict	Title
2026-04-28 14:25 UTC	INCONCLUSIVE	Sprague-Grundy Game Value of CNF Bounds Tree-Frege Lines
2026-04-28 14:45 UTC	INCONCLUSIVE	Roe-Skeleton Spectral Gap Forces Multiplicity in Protocol-Induced
2026-04-28 14:54 UTC	INCONCLUSIVE	Walsh-Code Antichain Width of NW Designs Bounds PRG Linear Bias
2026-04-28 15:27 UTC	INCONCLUSIVE	Kruskal-Katona Shadow Defect Lower-Bounds Monotone-KW Depth

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```